CROSS-CHAPTER CONSISTENCY AND TRACEABILITY

> **What this file is.** The required consistency matrix between the earlier chapters (One to Three) and the evidence, the traceability chains that connect them to Chapters Four to Six, and the contradictions that need the author's decision. Repository `student-beneficiary-system`, branch `fix/sbms-completion-work`, commit `d6b5032`.
>
> **Limitation on the left-hand column.** Chapters 1–3 are **not in this repository**. Their content is represented here by the approved proposal (`Project Proposal- 169613.pdf`) as transcribed in `SBMS_REQUIREMENTS_AND_SCOPE_CATALOGUE.md` §2 and by the cross-chapter checks already recorded in `EVIDENCE_AND_VERIFICATION_REPORT.md` §9. Any row marked **NOT INDEPENDENTLY VERIFIABLE** means the original wording could not be re-read here, not that it is wrong.

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1. Consistency matrix

Element	Original documentation	Verified implementation / evidence	Consistency status
Problem statement	Manual registration and tracking of beneficiaries is slow, error-prone and permits duplicate beneficiaries	Registration, screening, adjudication and payment-tracking modules all exist and are tested; the duplicate problem is addressed within a single registry	**CONSISTENT** — with the scope limit that cross-registry duplicates are out of scope (§1 of the catalogue)
General objective	Digitise registration, identification, verification and bursary tracking, with ML-assisted duplicate detection under human review	Three-tier system, 45 endpoints, 13 tables, advisory ML score, mandatory administrator decision	**CONSISTENT**
Specific objectives (i)–(iv)	Investigate challenges; design architecture, schema and ML pipeline with a Random Forest; implement registration, identifiers, QR and RBAC; validate and evaluate	Objective (i) as analysis; (ii) delivered with **XGBoost over six features**; (iii) delivered; (iv) partly delivered	**DRIFT on (ii)** — classifier altered; **PARTIAL on (iv)** — no real data, no UAT, no load test
Project scope	One registry, three roles, QR identification, awards, payments, certificates, reports	All present; six in-scope items remain gaps (FR-34…FR-37) and three proposal items are carried as deviations (FR-31…FR-33)	**CONSISTENT with declared exceptions** — every exception is numbered, none is silent
Functional requirements	Proposal §1.3 feature list, expanded into 37 numbered requirements in this package	30 delivered (24 verified, 6 partial), 3 deviations, 4 gaps	**CONSISTENT** — the numbering is this package's, and it maps back in `SBMS_REQUIREMENTS_AND_SCOPE_CATALOGUE.md` §6
Non-functional requirements	Performance, security, usability, maintainability, availability expectations	21 numbered NFRs; 14 verified, 7 partial; no load test, no usability study	**CONSISTENT with 7 partials** — the partials are named, not averaged away
Methodology	Design Science Research with Agile delivery (proposal §3.2–§3.3)	Artefact-centred build with iterative test-driven increments, evidenced by commit history and test growth (21/128 → 24/151)	**CONSISTENT** — the methods chapter's claim is supported by the build record, not contradicted by it
Technology choices	React frontend, Node/Express API, PostgreSQL, Python ML, Azure deployment	Exactly these (`frontend/`, `backend/`, `ml-service/`, `docs/deployment.md`, Azure logs)	**CONSISTENT**
ML methodology	Random Forest over four features (`jaro_winkler`, `dob_match`, `school_match`, `guardian_match`)	XGBoost over six features: the four above **minus** guardian-only framing plus `upi_match` and `birth_certificate_match`; the Random Forest survives only as a research candidate	**DRIFT** — must be stated in Chapters 4, 5 and 6 (G-21)

1.1 Where the ML row needs care

The delivered feature set is `jaro_winkler`, `dob_match`, `school_match`, `guardian_match`, `upi_match`, `birth_certificate_match` — six features, of which three are the proposal's, one is a reformulation (`guardian_match`

now covers two guardian fields), and two are new identity signals. The research artifact `ml-service/model/research/rf_attachment_style_candidate.joblib` is a ten-feature Random Forest and is **not** the production model. A writer who mixes the two produces a contradiction that a supervisor can find in one page.

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2. Traceability chains

Chain	Link 1 (Chapter 1–3)	Link 2 (Chapter 4)	Link 3 (Chapter 5)	Link 4 (Chapter 6)
Objective (i)	Proposal §1.1–§1.2, problem and challenges	§4.1–§4.2 requirement analysis	§5.5.2 evaluation of the objective	§6.2.3
Objective (ii)	Proposal §1.3.1 ii, §3.2.3	§4.4, §4.6, §4.7, §4.3.11	§5.3.1–§5.3.4	§6.2.4
Objective (iii)	Proposal §1.3.1 iii–iv	§4.2.2 (FR-01…FR-30), §4.4.2–§4.4.3	§5.3.2–§5.3.8, §5.4.3	§6.2.5
Objective (iv)	Proposal §1.3.1 v	§4.2.3 (NFR-01…NFR-21)	§5.4, §5.5, §5.6	§6.2.6–§6.2.7
Requirements	Proposal feature list	37 FRs and 21 NFRs, numbered in this package	Test coverage per requirement (`SBMS_TEST_EVIDENCE_CATALOGUE.md` §3–§6)	§6.2.7 and §6.2.9
Contribution	Proposal §1.4 (expected contribution)	§4.4.4 human-in-the-loop design	§5.5.5 ML evaluation and its limits	§6.2.8

`OBJECTIVE_REQUIREMENT_TEST_TRACEABILITY.md` carries the same chains at requirement granularity, including the NFR table and the per-suite coverage matrix; this section is the chapter-level view.

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3. Contradictions and drift requiring the author's confirmation

#	Item	What contradicts what	What the author must do
C-1	Classifier	Proposal: Random Forest, four features. Delivered: XGBoost, six features	Confirm the wording of the deviation in Chapters 4, 5 and 6 (already drafted) and in Chapter 3 if that chapter names the classifier
C-2	Deployment architecture	Proposal/Chapter 3 describe a containerised ML service; the delivered path also runs the model as a subprocess inside the API host	State both paths, and which one the Azure deployment uses (G-22)
C-3	`docs/verification.md`	Claims no controller calls the predictor; `students.controller.js:75` calls `scorePairs`	Mark the file superseded, or exclude it from the evidence base (G-08)
C-4	`docs/uat-test-plan.md`	Reports five `PASS` results with no participants, and one sentence that contradicts the implementation	Reclassify as an author walkthrough, or run a study (G-06, G-07)
C-5	Test totals	`docs/FINAL_TEST_REPORT.md` contains both 21/128 (2026-10-31) and 24/151 (2026-11-01)	Quote counts with their dates only (G-25)
C-6	Objectives wording	The exact Chapter 1 sentences could not be re-read here	Paste the originals into Chapter 6 and check §6.2.3–§6.2.6 answer them verbatim, not by paraphrase

No contradiction was resolved silently: each of the six is either stated in the drafts as a deviation, excluded from the evidence base, or listed here for decision.

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4. Naming consistency used across the package

Name used	Meaning	Do not substitute
`admin`, `field_officer`	The two role values in the code and database	Administrator/Field Officer only in prose, never as identifiers
`students`, `duplicate_flags`, `duplicate_review_events`	Tables	Never `beneficiaries` for the primary table

Name used	Meaning	Do not substitute
`bursary_awards`, `bursary_award_verification_events`	Award lifecycle	Never `awards` alone in a schema reference
`disbursements`, `disbursement_reconciliation_events`	Payment lifecycle	Never `payments` for the table
`bursary_certificates`	Certificate snapshot	Never `certificates` for the table
`audit_logs`, `reports`	Audit and reporting	—
`beneficiaries` (legacy)	Five legacy endpoints, kept for compatibility	Not part of the current design narrative

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5. Related files

`OBJECTIVE_REQUIREMENT_TEST_TRACEABILITY.md` (requirement-level traceability), `EVIDENCE_AND_VERIFICATION_REPORT.md` §9 (the checks behind the matrix), `SBMS_REQUIREMENTS_AND_SCOPE_CATALOGUE.md` §2 and §6 (proposal transcription and the FR register), `ACADEMIC_COMPLETENESS_AND_GAP_AUDIT.md` §4 (the same contradictions in gap form).